



EnzHier: Accurate Enzyme Function Prediction Through Multi-scale Feature Integration and Hierarchical Contrastive Learning

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Abstract. Accurate enzyme function prediction is essential for enzyme design and discovery. Existing methods face challenges with understudied enzyme families and multifunctional enzymes. We present EnzHier, a machine learning model that combines multi-scale feature integration with hierarchical triplet loss to predict Enzyme Commission (EC) numbers. By leveraging the hierarchical structure of EC classifications and multi-scale sequence similarity, EnzHier captures both local motifs and global sequence patterns. EnzHier outperforms state-of-the-art methods by achieving a 23% higher F1-score on benchmark cross-validation, and exhibits superior generalizability in external validations. The model also has high performance with challenging cases, correctly classifying difficult halogenases and identifying multifunctional enzymes like farnesyl pyrophosphate synthase—areas where other models often fail. Overall, EnzHier provides a robust and interpretable tool for enzyme function prediction, particularly for solving challenging cases where previous methods have shown limitations. EnzHier is publicly available at <https://github.com/labxscut/EnzHier>.

Keywords: Enzyme Function Prediction · Hierarchical Triplet Loss · Contrastive Learning · Multi-Scale Feature Integration

1 Introduction

Accurate enzyme function prediction is crucial for advancing enzyme design and discovery, and has far-reaching implications in food science, drug development,

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and other areas of biotechnology [1,2]. The Enzyme Commission (EC) numbering system [3], a four-level hierarchical scheme, encapsulates current knowledge of catalytic function. By grouping enzymes according to the reactions they catalyze, the EC system clarifies functional relationships within biological systems and supports applications in biochemistry, pharmacology, and industrial biotechnology. However, predicting enzyme function as an EC number remains an enduring challenge, particularly for enzymes from understudied families or those exhibiting multifunctional activities, with a wide gap of such knowledge in databases [4–7].

Traditional sequence homology-based methods (e.g. Blast [8,9]) remain widely used but exhibit significant limitations. Because accurate sequence-based function prediction demands high sequence identity ($\geq 50\text{--}60\%$) [10], these approaches suffer three critical drawbacks: they often misclassify when sequence similarity is moderate, lack adequate reference data for understudied enzymes, and fail to capture complex relationships in multifunctional enzymes when similarity signals are ambiguous. These constraints underscore the need for methods that extend beyond simple sequence similarity metrics.

To overcome these drawbacks, supervised learning approaches leveraging embedded sequence features have emerged as effective alternatives. Recognized methods such as HDMLF [11], DeepEC [12], DeepECTransformer [13], among others [14–16], exploited deep learning architectures like transformers and graph learning. Although these methods provide greater scalability and predictive power than sequence similarity based methods, they still rely heavily on large, well-annotated datasets, which are often limited for many enzymes. Consequently, their generalizability remains constrained, particularly for enzymes from underexplored families or those exhibiting multiple catalytic activities.

More recently, self-supervised learning techniques [17–19], including autoregressive [20] and contrastive learning [21,22], have emerged as a powerful alternative. Models such as CLEAN [21] use contrastive objectives to differentiate enzyme functions via positive and negative sample pairs [23]. Although these methods show enhanced performance on imbalanced datasets and understudied enzymes, they still fail to predict fine-grained enzyme functions or multifunctional enzymes accurately, because they do not explicitly model the EC hierarchy's subtle, multi-level relationships [24]. This shortcoming has limited their generalizability, a trend especially clear on external benchmarks like New-392 and Price-149, where CLEAN's performance drops significantly.

The fundamental limitation of existing contrastive learning approaches lies in their treatment of enzyme functions as flat, independent categories within the embedding space. For instance, consider three enzymes: EC 2.5.1.1 (dimethylallyltransferase), EC 2.5.1.10 (farnesyltransferase), and EC 1.1.1.86 (ketol-acid reductoisomerase). Standard contrastive learning methods position these enzymes equidistantly in the embedding space, failing to reflect their hierarchical functional relationships. A hierarchical approach would recognize that EC 2.5.1.1 and EC 2.5.1.10 both catalyze prenyl group transfers and differ only at the fourth EC level, indicating substrate specificity, whereas EC 1.1.1.86 differs

at the second level, representing a distinct reaction class. By adjusting the margin boundaries in the embedding space to account for EC-level similarities, such that enzymes sharing more EC levels are positioned closer together, the model can more accurately capture the nuanced relationships inherent in enzyme functions. This information about EC hierarchy is found to be effective in improving model performance with supervised approaches [11, 14]

Following this rationale, we integrate information from the EC hierarchy into a triplet contrastive loss [25], dynamically adjusting the distance margins in the embedding space based on the hierarchical similarity of the enzyme (Fig. 1). Our approach preserves the biochemical relationships within the EC hierarchy, enabling more accurate predictions in areas where conventional methods typically fail. Based on that, we introduce EnzHier, a novel machine learning model that outperforms existing methods on benchmark datasets. EnzHier performs particularly well in challenging cases, correctly identifying all functions of GGPP synthase and achieving superior prediction accuracy for difficult halogenase families. In a related independent work, hierarchical multilabel contrastive learning was found to be effective in predicting enzyme function, alphabet with smaller datasets [26].

2 Methods

2.1 Benchmark

We used 227,362 protein sequences from the SwissProt database [27], reserving 20% (45,472 sequences) as a test set and using 80% (181,890 sequences) for training. To prevent data leakage, we require at most 40% sequence similarity between the kept-out set and the training set. Two independent datasets served for external validation: NEW-392 (14.8% multifunctional enzymes) and Price-149 (2.01% multifunctional enzymes) [28].

Enzymes were categorized into common (>100 sequences) and rare (<100 sequences) classes, and we constructed a dedicated dataset of multifunctional enzymes carrying multiple EC numbers. This stratification enabled evaluation of EnzHier’s performance on both well-represented and understudied families and analysis under varying data sparsity conditions, focusing on multifunctional enzyme prediction accuracy. For comprehensive benchmarking, we compared EnzHier against state-of-the-art methods including CLEAN [21], DeepEC [12], ECPred [29], and BLAST-based approaches [9].

2.2 Hierarchical Triplet Loss

We developed a novel hierarchical triplet loss (HTL) to optimize enzyme embeddings representing the hierarchical nature of EC numbers. This function incorporates normalized similarity between EC numbers, weighted by their respective hierarchical levels (Fig. 1c). Specifically, for each anchor–positive–negative enzyme triplet, the HTL function $\mathcal{L}$ consists of the squared embedding distance between the anchor $z_a^{(i)}$ and positive $z_p^{(i)}$, the squared distance between the

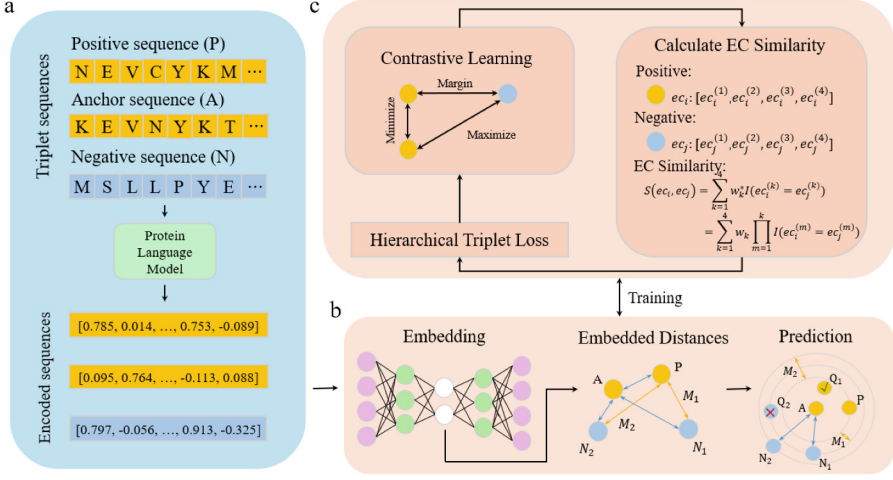


Fig. 1. EnzHier hierarchical contrastive learning framework. **(a)** Protein language models encode amino acid sequences into embedding vectors for contrastive learning. Triplets—anchor (A), positive (P) (sharing the same EC number with the anchor), and negative (N) (not sharing the same EC number with the anchor)—are marked in different colors. **(b)** Anchor-to-negative margins in the embedding space (such as M_1 and M_2) are dynamically expanded through contrastive learning, assigning hierarchical boundaries for EC levels and enabling subtle functional separation. **(c)** The final EnzHier model minimizes anchor-to-positive distances while maximizing anchor-to-negative distances. In the illustration, query sequences Q_1 and Q_2 are assigned to or excluded from the anchor’s EC group by comparing their distances (to the anchor’s EC center) against the learned margins.

anchor and negative $z_n^{(i)}$, and a dynamic margin m_{an} . $\mathcal{L}$ is minimized during training so that the model effectively separates positive from negative examples. To enforce non-negativity, we also apply a ReLU activation [30] to the HTL. In detail:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \max(0, \|z_a^{(i)} - z_p^{(i)}\|_2^2 - \|z_a^{(i)} - z_n^{(i)}\|_2^2 + m_{an}) \quad (1)$$

Here, N denotes the batch size. The dynamic margin m_{an} is adjusted according to the hierarchical similarity between EC labels. It is computed as follows:

$$m_{an} = m_{\min} - (m_{\min} - m_{\max}) \exp(-\beta S(ec_a, ec_n)) \quad (2)$$

In this expression, $m_{\min}$ and $m_{\max}$ are the minimum and maximum margin values, and β is a hyperparameter controlling the margin’s sensitivity to similarity. The similarity for the anchor–positive pair is set to $S_{ap} = S(ec_a, ec_p) = 1$, while $S_{an} = S(ec_a, ec_n)$ is the hierarchical similarity for the anchor–negative pair, as defined in Equation (3).

2.3 Hierarchical EC Similarity

To quantify hierarchical similarity between enzyme functions, we define a measure for EC numbers that honors their strict, nested relationships. Let ec_i and ec_j denote the EC numbers of enzymes i and j , each comprising four ordered levels from coarse to fine:

$$ec_i = [ec_i^{(1)}, ec_i^{(2)}, ec_i^{(3)}, ec_i^{(4)}], \quad ec_j = [ec_j^{(1)}, ec_j^{(2)}, ec_j^{(3)}, ec_j^{(4)}].$$

The similarity is defined as:

$$S(ec_i, ec_j) = \sum_{k=1}^4 w_k \prod_{m=1}^k I(ec_i^{(m)} = ec_j^{(m)}) \quad (3)$$

where

$$I(ec_i^{(m)} = ec_j^{(m)}) = \begin{cases} 1, & \text{if the } m\text{th EC field of } i \text{ and } j \text{ is identical,} \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

For each enzyme pair, $\prod_{m=1}^k I(\cdot)$ equals 1 only when all levels $1, \dots, k$ match; it drops to 0 if any level $m \leq k$ differs. Thus, $S \in [0, \sum_{k=1}^4 w_k]$. By normalizing $\sum_{k=1}^4 w_k = 1$, we ensure $S \in [0, 1]$, with higher scores indicating closer functional relatedness.

To reflect biological significance, one shall set the weights to decrease with depth: $w_1 > w_2 > w_3 > w_4$. We chose $w = (0.4, 0.3, 0.2, 0.1)$, i.e. $w_k \propto \{4, 3, 2, 1\}$. This makes a discrepancy at level 1 (broad reaction class) contribute more to the overall similarity than a mismatch at level 4 (substrate specificity), aligning with the EC hierarchy’s biological meaning. The values can also be fine-tuned by cross-validation.

Overall, the similarity measure remains simple, interpretable and computationally efficient; by assigning larger weights to higher (coarser) levels, it preserves the nested EC hierarchy while quantifying functional relatedness in a biologically meaningful way.

2.4 Multi-scale Feature Integration Module

To capture multi-scale information in protein sequences, we implement a modified U-Net architecture for 1D sequence processing with layer normalization. This architecture’s symmetric encoder–decoder design with skip connections effectively integrates detailed local features with broader contextual patterns.

The encoder pathway transforms input sequence features $X \in \mathbb{R}^{L \times d_{\text{in}}}$ (L : sequence length; $d_{\text{in}} = 2560$ from ESM-2 embedding) through sequential dimensionality reduction, where each layer downsamples by halving, as follows:

$$H_1 = \text{ReLU}(\text{LayerNorm}(X W_1 + b_1)) \cdot M_1 \quad (5)$$

$$H_2 = \text{ReLU}(\text{LayerNorm}(H_1 W_2 + b_2)) \cdot M_2 \quad (6)$$

$$H_3 = \text{ReLU}(\text{LayerNorm}(H_2 W_3 + b_3)) \cdot M_3 \quad (7)$$

The decoder pathway reverses this reduction while incorporating encoder features through skip connections:

$$D_1 = \text{ReLU}(\text{LayerNorm}(H_3 W_4 + b_4)) \cdot M_4 \quad (8)$$

$$D_1^{\text{concat}} = [D_1 \oplus H_2] \quad (9)$$

$$D_2 = \text{ReLU}(\text{LayerNorm}(D_1^{\text{concat}} W_5 + b_5)) \cdot M_5 \quad (10)$$

$$D_2^{\text{concat}} = [D_2 \oplus H_1] \quad (11)$$

$$\hat{Y} = D_2^{\text{concat}} W_{\text{out}} + b_{\text{out}} \quad (12)$$

Here $\hat{Y} \in \mathbb{R}^{L \times d_{\text{out}}}$ and $d_{\text{out}} = 2560$ is the output feature dimension. These skip connections ($D_1 \oplus H_2$ and $D_2 \oplus H_1$) preserve fine-grained local information from the encoder while incorporating global contextual features from the decoder, essential for distinguishing subtle functional differences between enzymes with similar sequences but distinct catalytic activities.

2.5 EC Number Inference Method

We employ a change-point detection inference to assign EC numbers by detecting large gaps in the sorted distances between a query and class centroids. Let E be the total number of EC classes, indexed $j = 1, 2, \dots, E$. Denote by c_j the centroid embedding of class j and by q the query embedding. Compute

$$d_j^q = \|q - c_j\|_2, \quad j = 1, \dots, E.$$

To focus on likely candidates, let k (default 15) be the maximum number of top classes to consider, and define the candidate index set:

$$\mathcal{C} = \{j : d_j^q \leq d_{(k)}^q\}, \quad |\mathcal{C}| = k. \quad (13)$$

Within the set $\mathcal{C}$, compute adjacent gaps:

$$\Delta_i = d_{(i+1)}^q - d_{(i)}^q, \quad i = 1, \dots, k-1.$$

Identify the change point i^* where Δ_{i^*} is largest (and exceeds the mean plus twice the standard deviation of $\{\Delta_i\}$), indicating a natural separation between close and distant classes. Assign the query to all EC classes $j \in \mathcal{C}$ with $d_j^q \leq d_{(i^*)}^q$.

DBSCAN Fallback: Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is our alternative for grouping EC centroids that are closely packed into dense regions [31]. Suppose gap-based changepoint detection fails to produce a clear separation. In that case, we apply one-dimensional DBSCAN to the distances $\{d_j^q : j \in \mathcal{C}\}$ requiring a cutoff $\epsilon = 0.5 \sigma_d$ and a minimum one point per cluster, where σ_d is the standard deviation of these d^q 's. Centroid clusters identified by DBSCAN correspond to groups of nearby centroids; the query is assigned to the EC class(es) linked to the cluster containing $d_{(1)}^q$.

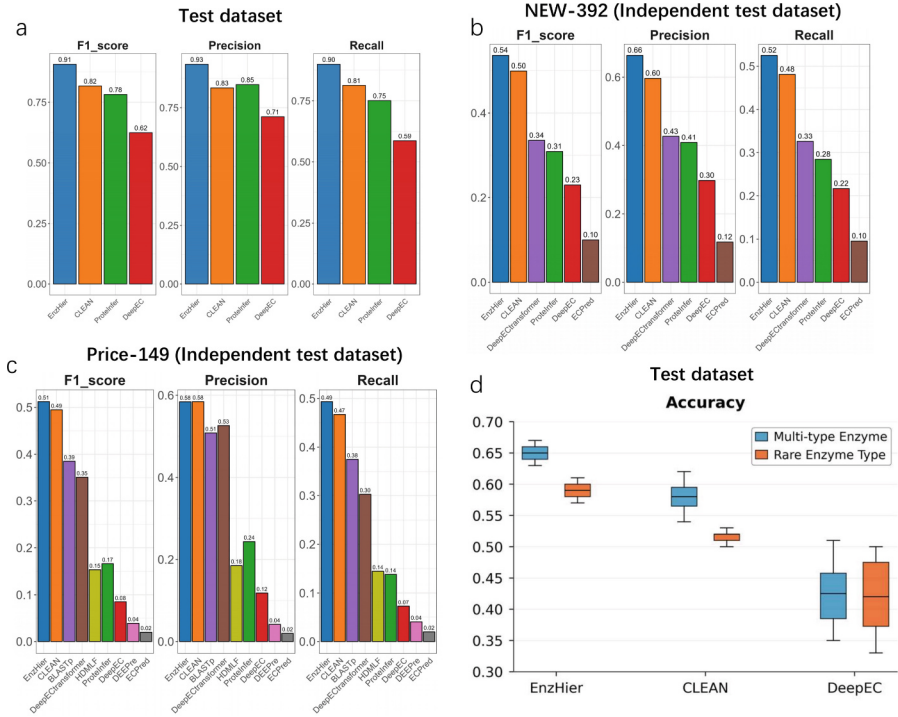


Fig. 2. The performance (including Precision, Recall, and F1-score) of EnzHier and other methods on (a) the keptaway test dataset, (b) the New-392 independent dataset, and (c) the Price-149 independent dataset. (d) compares the prediction accuracy of EnzHier, CLEAN, and DeepEC in multi-type and rare enzyme subsets.

3 Results

3.1 EnzHier Achieves State-of-the-Art Performance in Benchmarks

EnzHier demonstrates robust and consistent performance across 5-fold cross-validation (Supplementary Table S3), achieving an average precision of 0.943, recall of 0.914, F1 score of 0.919, and AUC of 0.957. On the held-out test set (20% of SwissProt), EnzHier outperforms all comparison methods—achieving an F1 of 0.91, precision of 0.93, and recall of 0.90—surpassing CLEAN, ProtInfer, and DeepEC (Fig. 2a). These results highlight EnzHier’s balanced precision–recall trade-off in enzyme function prediction.

On the independent NEW-392 dataset, EnzHier outperforms all other methods across F1, precision, and recall (Fig. 2b). CLEAN falls behind in every metric, and ProtInfer, DeepEC, and ECpred underperform across all measures. Similarly, on Price-149, EnzHier achieves the highest F1, precision, and recall (Fig. 2c). Although CLEAN’s precision is closer to EnzHier’s, its recall and F1 are all lower; ProtInfer and DeepEC degrade further, and ECpred scores lowest.

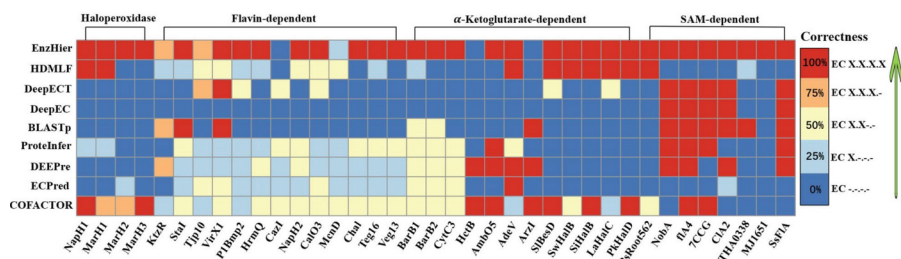


Fig. 3. Prediction correctness for halogenase classification. The heatmap shows prediction correctness (0% to 100%) for haloperoxidase, flavin-dependent, α -ketoglutarate-dependent, and SAM-dependent halogenase subtypes based on EC number matching. Red indicates complete matching (100%), followed by orange, yellow, light blue, and dark blue (75%, 50%, 25%, and 0% matching, respectively). (Color figure online)

EnzHier also works well on understudied enzyme classes. Figure 2d shows accuracy distributions for multi-type and rare enzyme test subsets. EnzHier attains median accuracies of approximately 0.65 for multi-type and 0.59 for rare classes, outperforming CLEAN (0.58 and 0.53) and DeepEC (0.43 and 0.42). All methods trend higher on multi-type enzymes than on rare types, but EnzHier’s improvement is most pronounced.

3.2 EnzHier Accurately Predicts Halogenase Functions

EnzHier demonstrates exceptional accuracy in predicting complex and difficult-to-classify enzymes, such as halogenases. It has high performance in classifying all four halogenase subtypes—haloperoxidases, flavin-dependent, α -ketoglutarate-dependent, and SAM-dependent halogenases, achieving high (>90%) average correctness for all subcategories (Fig. 3). By contrast, HDMLF attains high accuracy only on a few haloperoxidases but frequently misclassifies SAM- and flavin-dependent halogenases and shows inconsistent performance on α -ketoglutarate-dependent subtypes. DeepEC and DeepECTransformer similarly misclassify large fractions of flavin- and α -ketoglutarate-dependent halogenases and fail significantly on haloperoxidases. All other methods also demonstrate significant limitations in predicting halogenase functions.

3.3 EnzHier Effectively Annotates Multifunctional Enzymes

EnzHier also excels at annotating multifunctional enzymes—a task where existing methods often struggle. A prime example is geranylgeranyl pyrophosphate synthase (GGPP synthase; UniProt ID: B1XJV9), shown in Supplementary (Fig. 3). Encoded by the *CRTE* gene, this 302-amino-acid enzyme sequentially adds three isopentenyl pyrophosphate molecules to dimethylallyl pyrophosphate, producing GGPP—a critical step in cyanobacterial terpenoid biosynthesis confirmed by crystal structure. UniProt currently assigns GGPP synthase to three EC categories:

EC 2.5.1.1, EC 2.5.1.10, and EC 2.5.1.29. EnzHier correctly recovers all three functional annotations, whereas DeepECTransformer and DeepEC capture only a subset and DEEPRe fails entirely. This highlights EnzHier’s superior performance in annotating multifunctional enzymes.

3.4 Ablation Studies

Table 1. Performance of Ablated Model Variants

Model Variant	Precision	Recall	F1-Score	AUC
Base (Flat T-Loss + CNN)	0.8245	0.8102	0.8173	0.8956
+ Hierarchical Triplet Loss	0.9021	0.8743	0.8880	0.9312
+ U-Net Architecture	0.9094	0.8891	0.8877	0.9287
EnzHier	0.9132	0.9038	0.8994	0.9469

To evaluate the contribution of each component of EnzHier to overall performance, we conducted comprehensive ablation studies. First, we compared various protein language models—ESM-1b and ESM-2 across parameter scales from 8 million up to 3 billion. While larger models provided gains, those improvements plateaued beyond 150 million parameters, leading us to select ESM-2 (150 M) for an optimal performance-efficiency trade-off.

Next, we replaced the standard CNN architecture with our modified U-Net (Table 1). This “encoder–decoder with skip connections” design outperformed CNN (by 2.85%), Transformer (by 1.33%), and ResNet (by 1.83%) across all metrics. U-Net’s ability to capture local motifs (via convolutions) and global context (via skip links) makes it particularly effective in distinguishing subtle sequence features relevant to enzyme function.

Finally, switching from flat triplet loss (Flat T-Loss) to hierarchical triplet loss (HTL) provided the single largest boost: F1 score increased by 7.07% (from 0.8173 to 0.8880). Incorporating U-Net alone improved F1 by 7.04% (to 0.8877). Combining HTL and U-Net yielded the highest F1 (0.8994; an 8.21% jump) and the highest AUC (0.9469; a 5.13% increase). Repeating training with different random seeds produced consistent results ($\pm 1.2\%$ variation), confirming the robustness of these design choices.

4 Conclusion

In summary, EnzHier advances enzyme function prediction with improved accuracy, robustness, and scalability. It effectively predicts functions of understudied and multifunctional enzymes like halogenases and geranylgeranyl pyrophosphate synthase, addressing key challenges in enzyme engineering. The model combines

contrastive learning with hierarchical triplet loss and U-Net architecture for optimal feature extraction. With its high performance and versatility, EnzHier could be a valuable tool added for enzyme function prediction.

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